



FINAL-TERM EXAMINATION

Spring 2026

COURSE CODE	CSC-201	COURSE TITLE	Data Structures and Algorithms
PROGRAMME	BS Computer Science	SEMESTER	III
CREDIT HOURS	3 (3-0)	DATE	16 June 2026
TIME ALLOWED	3 hours	TOTAL MARKS	50
VENUE	Examination Hall 2		

ROLL NO. _____ NAME _____ SIGNATURE _____

INSTRUCTIONS

- Attempt **all** questions. Section A is answered on this paper; Sections B and C in the answer book provided.
- The marks for each question are shown in the margin, with the course learning outcome it assesses.
- Programmable calculators, mobile telephones and smart watches are not allowed in the examination hall.
- Write your roll number on every sheet of the answer book.

FOR THE EXAMINER ONLY

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	TOTAL
MARKS AWARDED													

Examiner's signature _____

Date _____

SECTION A: MULTIPLE CHOICE

10 marks

- Q1.** What is the worst-case time complexity of binary search on a sorted array of n elements?
- (a) $O(1)$ (b) $O(\log n)$ (c) $O(n)$ (d) $O(n \log n)$
- Q2.** Which structure serves its items in last-in, first-out order?
- (a) Queue (b) Stack
(c) Priority queue (d) Circular buffer

2 marks
CLO 1
Remember

2 marks
CLO 1
Understand



- Q3.** A binary search tree holding n keys has height h . Searching it takes time proportional to:
- (a) n in every case (b) h , which is $\log n$ only when the tree is balanced
(c) $\log n$ in every case (d) $n \log n$ in the worst case
- Q4.** Separate chaining resolves a collision by:
- (a) probing the next free slot (b) keeping a list of the entries that share a slot
(c) enlarging the table on every insertion (d) rehashing with a second function
- Q5.** Breadth-first search on an unweighted graph finds:
- (a) the minimum spanning tree (b) the shortest path in edges from the source
(c) the strongly connected components (d) a topological order

2 marks
CLO 1
Understand

2 marks
CLO 2
Apply

2 marks
CLO 2
Apply

SECTION B: SHORT QUESTIONS

20 marks

- Q6.** Write a recursive function that returns the height of a binary tree, and state its running time in terms of the number of nodes.
- Q7.** Complete the table below, and say which of the three you would choose for an array that is already nearly sorted.

5 marks
CLO 2
Apply

5 marks
CLO 2
Analyse

Algorithm	Best	Average	Worst
Insertion sort	$O(n)$		
Merge sort		$O(n \log n)$	
Quicksort			

- Q8.** The running time of a divide-and-conquer algorithm satisfies

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n), \quad T(1) = \Theta(1).$$

5 marks
CLO 3
Analyse

Solve the recurrence and name an algorithm whose running time it describes.

- Q9.** A campus application keeps the 815 enrolled students in memory and looks a student up by roll number thousands of times a minute, while enrolments are added only a few times a day. Recommend a data structure and justify the choice in terms of the operations' costs.

5 marks
CLO 3
Evaluate

**SECTION C: LONG QUESTIONS****20 marks**

Q10. Consider the array [38, 27, 43, 3, 9, 82, 10].

(a) Trace merge sort on it, showing the contents of the array after every merge.
[6 marks]

10 marks
CLO 3
Analyse

(b) Count the comparisons merge sort makes on this input and compare the number with insertion sort on the same input.

[4 marks]

Q11. A hash table stores the courses a department offers, keyed by course code.

(a) Write the insertion routine for separate chaining. One line of code per line of the answer book:

```
def insert(table, key, value):
```

[4 marks]

(b) The table has 16 slots and holds 40 courses. Explain what the load factor does to the cost of a lookup, and describe one way to keep the cost near constant as the department adds courses. Use the space below for a diagram.

[6 marks]

Q12. BONUS State one change to the hash table above that would make its iteration order predictable, and what it would cost.

3 marks
CLO 4
Create

END OF QUESTION PAPER



MARKS AND CLO DISTRIBUTION

QUESTION	TOPIC	CLO	LEVEL	MARKS
1	Complexity	CLO 1	Remember	2
2	Stacks and queues	CLO 1	Understand	2
3	Trees	CLO 1	Understand	2
4	Hashing	CLO 2	Apply	2
5	Graphs	CLO 2	Apply	2
6	Recursion	CLO 2	Apply	5
7	Sorting	CLO 2	Analyse	5
8	Complexity	CLO 3	Analyse	5
9	Data structure choice	CLO 3	Evaluate	5
10	Sorting	CLO 3	Analyse	10
11	Hashing	CLO 4	Create	10
12	Open question	CLO 4	Create	+3
TOTAL				50